

Code-Layout, Readability and Reusability

Date	01 July 2026
Team ID	N/A
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Uniform indentation and formatting followed across all Python files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into app, services, tests, static files, and configuration folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear variable names such as event_details, topics, summary, and conversation_history.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like generate_topics(), fact_check(), and analyze_event() clearly describe their purpose.
5	Code Comments	Inline and block comments explain log	Partial	Important functions are documented with comments where required.
6	Modular Design	Code is split into reusable functions/modules	Yes	Event Analyzer, Topic Generator, Fact Checker, and API routes are implemented as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions reduce duplicate code across services.
8	Error Handling	Exceptions and errors are handled carefully	Yes	API exceptions and invalid user inputs are handled gracefully.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Event Analyzer Service	Python, DistilBERT	Event classification	High

2	Topic Generator Service	Python, GPT-2	Conversation topic generation	High
3	Fact Checker Service	Python, Wikipedia API	Topic verification	High
4	FastAPI Routes	FastAPI	Backend API endpoints	High
5	Streamlit UI Components	Streamlit	User Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with modular architecture.
Readability	5	Easy-to-read code with meaningful naming conventions.
Reusability	5	Core services are reusable across multiple workflows.
Documentation / Comments	4	Adequate comments and project documentation provided.
Overall Score	4.8/5	Clean, modular, maintainable, and scalable codebase suitable for future enhancements.